

Computer Science & IT

Database Management System



Relational Model & Normal Forms

Lecture No. 08



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Recap of Previous Lecture



Topic

Membership test



Topic

Relation between two FD sets



Topic

FD set of a sub-relation



Topics to be Covered



- ✓ **Topic** Minimal cover or Canonical cover
- ✓ **Topic** Schema refinement (Normalization)
- ✓ **Topic** Problems because of redundancy in a relation
- ✓ **Topic** Properties of decomposition





Topic : Minimal cover (Canonical cover)

- Minimal cover of canonical cover of FD set F is a set of functional dependencies (F_m) such that,
- $F_m = F$ and
 - F_m does not contain any redundant FD, and F_m must not contain any extraneous attribute at either side of any of its FD
L.H.S.



Topic : Procedure to obtain minimal cover of FD set

1. Simplify RHS of all FDs (i.e., split the FDs such that RHS contain exactly one attribute)
2. For all FDs find redundant (extraneous) attribute in LHS.
3. Eliminate all redundant FDs
4. Apply Union if needed
5. The result is minimal Cover

{ As soon as any redundant FD is identified, eliminate that FD. }

{ As soon as any extraneous attribute is identified, we will remove that attribute and check other FDs }



Topic : Testing if an Attribute is Extraneous in LHS

Consider a set F of functional dependencies and functional dependency $\alpha \rightarrow \beta$ in F .

set of attributes

Set of attributes

To test if attribute $A \in \alpha$ is extraneous in α (i.e., Any extraneous attribute in LHS of FD)

1. compute $(\{\alpha\} - \{A\})^+$ using the dependencies in F' , where $F' = (F - \{\alpha \rightarrow \beta\})$
2. check if $(\{\alpha\} - A)^+$ contains A ; if it does then, A is extraneous.

#e.g., Consider the FD set

$$F = \{AB \rightarrow C, B \rightarrow A, A \rightarrow B\}$$

$$\left\{ \begin{array}{l} AB \rightarrow C \\ B \rightarrow A \\ A \rightarrow B \end{array} \right.$$

$$\underbrace{AB}_{\text{check if "A" is extraneous in (AB)}} \rightarrow C$$

$$\left((\{A, B\} - \{A\})^+ \text{ w.r.t. } (F - AB \rightarrow C) \right)$$

$$= B \quad = \{A, B\}$$

$$A \in (\{A, B\} - \{A\})^+ \text{ w.r.t. } F - (AB \rightarrow C)$$

∴ A is extraneous

∴ Eliminate A from L.H.S. of $AB \rightarrow C$
 ∴ FD becomes $B \rightarrow C$

after elimination of A from $AB \rightarrow C$
new FD set becomes

$$\left\{ \begin{array}{l} B \rightarrow C \\ B \rightarrow A \\ A \rightarrow B \end{array} \right\}$$

#e.g., Consider the FD set

$$F = \{AB \rightarrow \underline{C}, \underline{B} \rightarrow \underline{A}, \underline{A} \rightarrow \underline{B}\}$$

$$\left. \begin{array}{l} AB \rightarrow C \\ \checkmark \underline{B} \rightarrow A \\ \checkmark \underline{A} \rightarrow B \end{array} \right\}$$

$$\begin{array}{l} \underline{AB} \rightarrow C \\ \quad \downarrow \text{check if } B \text{ is extraneous in } (AB) \\ (\{A, B\} - \{B\})^+ \text{ w.r.t. } (F - (AB \rightarrow C)) \\ \quad = A \qquad \qquad \qquad = \{A, B\} \end{array}$$

- $B \in (\{A, B\} - \{B\})^+ \text{ w.r.t. } F - (AB \rightarrow C)$
- $\therefore B$ is extraneous
- $\therefore$ Eliminate B from L.H.S. of $AB \rightarrow C$
- $\therefore$ FD becomes $A \rightarrow C$

after elimination of B from $AB \rightarrow C$
new FD set becomes

$$\left\{ \begin{array}{l} A \rightarrow C \\ B \rightarrow A \\ A \rightarrow B \end{array} \right\}$$



Topic : Testing if any FD is redundant

Set of attributes

Consider a set F of functional dependencies and functional dependency $\alpha \rightarrow \beta$ in F .

To test if $\alpha \rightarrow \beta$ is redundant

1. compute α^+ using only the dependencies in F' , where $F' = (F - \{\alpha \rightarrow \beta\})$
2. check that α^+ contains β ; if it does then, $\alpha \rightarrow \beta$ is redundant

Single attribute

#e.g., Consider the FD set

$F = \{AB \rightarrow C, B \rightarrow A, A \rightarrow B\}$, Find minimal cover of FD set F .

We know in $AB \rightarrow C$, "A" is Extraneous if B is present
 & in $AB \rightarrow C$, B is extraneous if A is present

If we consider A is Extraneous in $AB \rightarrow C$
 then resulting set of FD

$F_1 = \left\{ \begin{array}{l} B \rightarrow C \\ B \rightarrow A \\ A \rightarrow B \end{array} \right\}$ } None of them is redundant

No Extraneous attribute in LHS,
 & No redundant FD in F_1
 $\therefore F_1$ is minimal cover of F

If we consider B is Extraneous in $AB \rightarrow C$
 then resulting set of FD

$F_2 = \left\{ \begin{array}{l} A \rightarrow C \\ B \rightarrow A \\ A \rightarrow B \end{array} \right\}$ } None of them is redundant

No Extraneous attribute in LHS of any FD
 & No redundant FD in set F_2 , $\therefore F_2$ is also minimal cover of F



Topic : NOTE



Minimal cover of FD set F need not be unique, but all minimal cover are logically equivalent.

Note: If F_{m_1} & F_{m_2} are two minimal covers of FD set F ,
then $F_{m_1} = F_{m_2}$

#e.g., Consider the FD set

$$F = \{A \rightarrow BC, B \rightarrow C\}$$

Find minimal cover of FD set F .

$$F = \left\{ \begin{array}{l} A \rightarrow BC \\ B \rightarrow C \end{array} \right\} \xrightarrow{\text{Simplify}} \left\{ \begin{array}{l} A \rightarrow B \\ A \rightarrow C \\ B \rightarrow C \end{array} \right\}$$

↑
No extraneous
attribute in
LHS.

$A \rightarrow C$ is
redundant
∴ Eliminate
 $A \rightarrow C$

$\Rightarrow$

$A \rightarrow B$
 $B \rightarrow C$
is Minimal
Cover of F

#e.g., Consider the following FD set

$F = \{AC \rightarrow G$

$D \rightarrow EG$

$BC \rightarrow D$

$CG \rightarrow BD$

$ACD \rightarrow B$

$CE \rightarrow AG$

$\}$

Find minimal cover of F .

Find minimal cover of

$$AC \rightarrow G$$

$$D \rightarrow EG$$

F: $BC \rightarrow D$

$$CG \rightarrow BD$$

$$ACD \rightarrow B$$

$$CE \rightarrow AG$$

Simplify w.r.t. RHS

$$AC \rightarrow G$$

$$D \rightarrow E$$

$$D \rightarrow G$$

$$BC \rightarrow D$$

$$CG \rightarrow B$$

$$CG \rightarrow D$$

$$ACD \rightarrow B$$

$$CE \rightarrow A$$

$$CE \rightarrow G$$

Eliminate Extraneous attributes from LHS, if any

$$(A)^+ = \{A\} \because C \text{ is not Extra} \therefore AC \rightarrow G$$

$$(C)^+ = \{C\} \because A \text{ is not Extra}$$

$$\text{only one attribute in LHS} \therefore D \rightarrow E$$

$$\text{only one attribute in LHS} \therefore D \rightarrow G$$

$$(B)^+ = \{B\} \because C \text{ is not Extra} \therefore BC \rightarrow D$$

$$(C)^+ = \{C\} \because B \text{ is not Extra}$$

$$(G)^+ = \{G\} \because G \text{ is not Extra} \therefore CG \rightarrow B$$

$$(C)^+ = \{C\} \because G \text{ is not Extra} \therefore CG \rightarrow D$$

$$(G)^+ = \{G\} \because G \text{ is not Extra}$$

$$(AC)^+ = \{A, C, G, B, D, \dots\}$$

$$(AD)^+ = \{A, D, E, G\}$$

$$(CD)^+ = \{C, D, E, G, A, \dots\}$$

$$(C)^+ = \{C\}$$

$$(E)^+ = \{E\}$$

$$\text{None of } C \text{ or } E \text{ is Extra}$$

$DE(AC)^+$
 $\therefore D$ is Extraneous w.r.t. (AC)

A is Extra w.r.t. (CD)
 C is not Extra w.r.t. AD

There are two Choices
either use $AC \rightarrow B$
or use $CD \rightarrow B$

We can further Compute

$$(C)^+ \rightarrow (A)^+$$

$$(C)^+ \rightarrow (D)^+$$

To know if there are more than one Extraneous attributes in LHS of $ACD \rightarrow B$

Find minimal cover of

$$AC \rightarrow G$$

$$D \rightarrow EG$$

F: $BC \rightarrow D$

$$CG \rightarrow BD$$

$$ACD \rightarrow B$$

$$CE \rightarrow AG$$

Simplify wrt RHS

$$AC \rightarrow G$$

$$D \rightarrow E$$

$$D \rightarrow G$$

$$BC \rightarrow D$$

$$CG \rightarrow B$$

$$CG \rightarrow D$$

$$ACD \rightarrow B$$

$$CE \rightarrow A$$

$$CE \rightarrow G$$

Eliminate Extraneous attributes from LHS (if any)

$(A)^+ = \{A\}$ $\therefore C$ is not Extra
 $(C)^+ = \{C\}$ $\therefore A$ is not Extra

only one attribute in LHS $\therefore$

only one attribute in LHS $\therefore$

$(B)^+ = \{B\}$ $\therefore C$ is not Extra
 $(C)^+ = \{C\}$ $\therefore B$ is not Extra

$(C)^+ = \{C\}$ $\therefore G$ is not Extra
 $(G)^+ = \{G\}$ $\therefore C$ is not Extra

$(C)^+ = \{C\}$ $\therefore G$ is not Extra
 $(G)^+ = \{G\}$ $\therefore C$ is not Extra

$(AC)^+ = \{A, C, G, B, D, \dots\}$
 $(AD)^+ = \{A, D, E, G\}$
 $(CD)^+ = \{C, D, E, G, A, \dots\}$
 } let's consider D as Extraneous attribute wrt (AC)

$(C)^+ = \{C\}$ None $\rightarrow$
 $(E)^+ = \{E\}$ all $\rightarrow$
 $\therefore C \text{ or } E$ is Extraneous

FD set with No Extra attribute in LHS.

$$AC \rightarrow G$$

$$D \rightarrow E$$

$$D \rightarrow G$$

$$BC \rightarrow D$$

$$CG \rightarrow B$$

$$CG \rightarrow D$$

$$AC \rightarrow B$$

$$CE \rightarrow A$$

$$CE \rightarrow G$$

Eliminate Redundant FD

$$(AC)^+ = \{A, C, B, D, G\}$$

FD set with No Extra attribute in LHS.

~~$AC \rightarrow G$~~

$D \rightarrow E$

$D \rightarrow G$

$BC \rightarrow D$

$AC \rightarrow B$

$CE \rightarrow A$

~~$CE \rightarrow G$~~

$(CG)^+ = \{C, G, B, D\}$
without $CG \rightarrow D$
 $\therefore CG \rightarrow D$ is Redundant

$CG \rightarrow D$

$CG \rightarrow B$

~~$CG \rightarrow B$~~

~~$CG \rightarrow D$~~

Eliminate Redundant FD

$(AC)^+ = \{A, C, B, D, \textcircled{G}\}$

$(D)^+ = \{D, G\} \therefore$ Not Redundant

$(D)^+ = \{D, E\} \therefore$ Not Redundant

$(BC)^+ = \{B, C\} \therefore$ Not Redundant

$(CG)^+ = \{C, G, D, E, A, \textcircled{B}\} \therefore$ Redundant

$(CG)^+ = \{C, G\} \therefore$ Not Redundant

$(AC)^+ = \{A, C\} \therefore$ Not Redundant

$(CE)^+ = \{C, E, G, D\} \therefore$ Not Redundant

$(CE)^+ = \{C, E, A, B, D, \textcircled{G}\} \therefore$ Redundant

We can say $AC \rightarrow G$ is redundant,
(or) we can say "G" is extraneous in R.H.S

Final set of FDs

$D \rightarrow E$

$D \rightarrow G$

$BC \rightarrow D$

$CG \rightarrow D$

$AC \rightarrow B$

$CE \rightarrow A$

It is one of the minimal cover of F.

#e.g., Consider the following FD set

$F = \{A \rightarrow BC$

$CD \rightarrow E$

$E \rightarrow C$

$D \rightarrow AEH$

$ABH \rightarrow BD$

$DH \rightarrow BC$

$\}$

Find minimal cover of F.



Topic : Schema refinement (Normalization)



Normalization is a process of reducing / eliminating the redundancy present in the relational table



Topic : Redundancy in a relation

If two or more independent informations are stored in a single table, then redundancy is possible.

(redundancy
↓
duplicated information)

٩٠

[illegible]

We are storing independent informations in the same table.

- o Redundancy exist.



Topic : Problems because of redundancy

- ① Insertion Anomaly
- ② Deletion Anomaly
- ③ Update Anomaly

H.W.

Read about them



2 mins Summary



✓
Topic

Minimal cover or Canonical cover

✓
Topic

Schema refinement (Normalization)

Topic

Problems because of redundancy in a relation

Topic

Properties of decomposition

THANK - YOU